

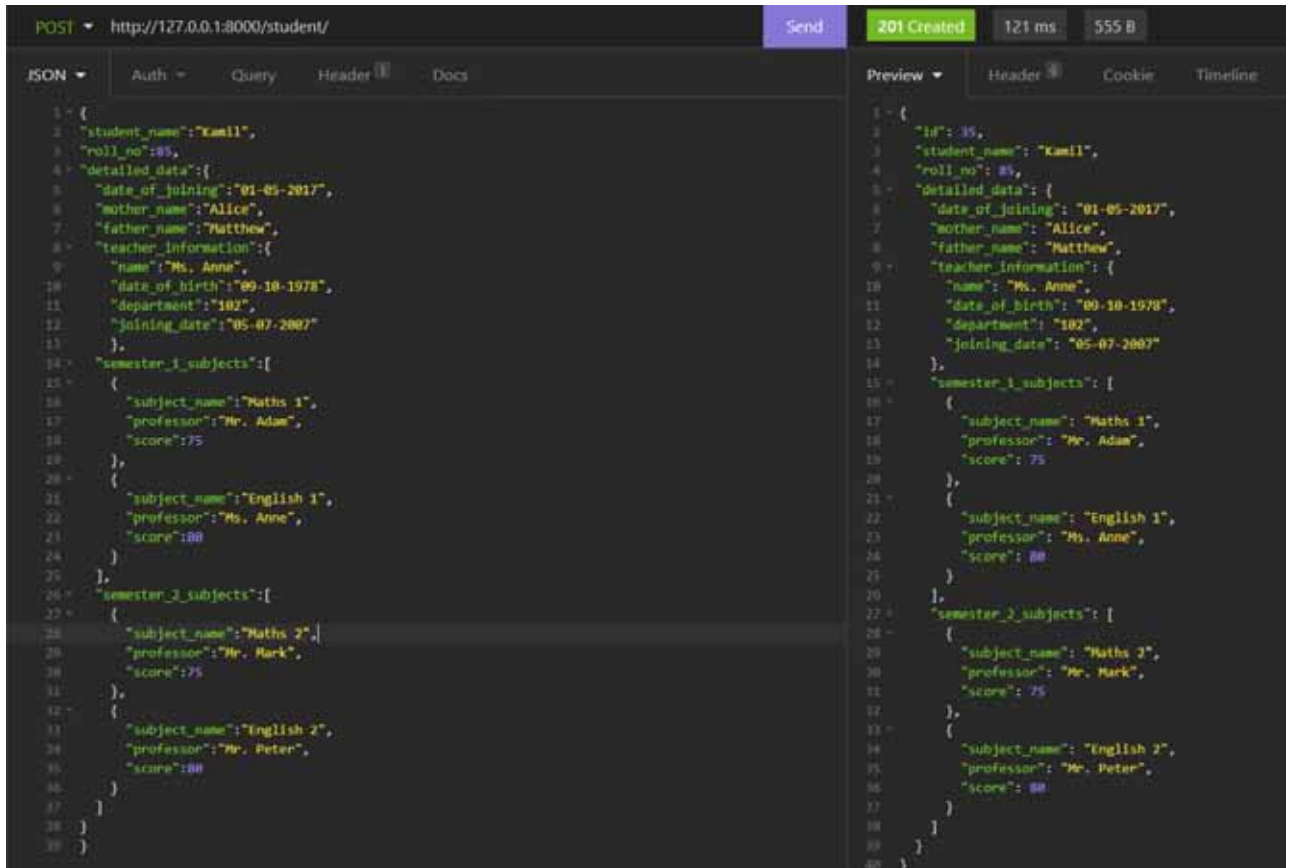


Figure 8: Successful validation of JSONField schema

```
myschema["required"].extend(["teacher_
information", "semester_1_subjects", "semester_2_subjects"])
myschema["properties"]["date_of_
joining"]["pattern"]='^[0-9]{2}-[0-9]{2}-[0-9]{4}$' ----(15)
myschema["properties"]
["teacher_information"]["properties"]["joining_date"]
["pattern"]='^[0-9]{2}-[0-9]{2}-[0-9]{4}$'
myschema["properties"]
["teacher_information"]["properties"]["date_of_birth"]
["pattern"]='^[0-9]{2}-[0-9]{2}-[0-9]{4}$'
v=Draft7Validator(myschema)
if len(list(v.iter_
errors(data["detailed_data"])))!=0:
    return
Response({"error":str(list(v.iter_errors(data["detailed_
data"]))))} ----(20)
else:
    serializer.save()
    return Response(serializer.
data, status=status.HTTP_201_CREATED)
else:
    return Response(serializer.
errors, status=status.HTTP_400_BAD_REQUEST) ----(25)
```

The section of views from lines (10) to (13) aims to convert the structure of 'semester_1_subjects' and 'semester_2_subjects' from 'type':'object' to 'type':'array'. Line (14) intends to append 'teacher_information', 'semester_1_subjects' and 'semester_2_subjects' to the outermost 'required' list. Lines (15) to (17) help include a 'pattern' key-value pair in the schema structure for date-related fields.

So you now have a basic understanding of how the structure of data of a JSONField within a Django REST framework model can be cross-checked against a well-defined schema. Covering the lengthier method, i.e., writing the schema yourself, was essential to understand how a schema looks. You also have an idea of how JSON models help generate the schema for you, and even though the auto-generated schema is not an exact reflection of the schema you may write yourself, you can manipulate it to suit your needs utilising a fundamental data structure in Python: dictionaries, and its basic operations. **END** 🐧

By: Mohammed Kamil Khan

The author is currently working at Caravel.AI. His aim is to make available the knowledge of topics he comes across in a beginner-friendly manner.